

Xijing Wang

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Education

Carnegie Mellon University, School of Computer Science

M.S. in Automated Science | Director's Fellowship

Pittsburgh, PA

Aug 2025 – May 2027

Santa Clara University, College of Arts and Sciences

B.S. in Computer Science (Algorithms), Minor in Economics | REAL Program Scholar

Santa Clara, CA

Sep 2021 – Mar 2025

Technical Skills

Languages: Python, C++, Swift, Java, Go, TypeScript/JavaScript, SQL

Backend / Systems: FastAPI, REST APIs, SQLAlchemy, PostgreSQL, SQLite, Redis, Docker, Linux/Shell, Git, Pytest

Frontend / Mobile: SwiftUI, UIKit, React Native, React, Redux, **Core ML**, HealthKit

AI Infrastructure: LLM Agents, Agentic Workflows, **RAG**, **coremltools**, **MLX**, **MLC-LLM**, **TVM**, PyTorch

Experience

Apple

Software Engineer Intern | Python, Core ML, coremltools, MLX, Swift

Cupertino, CA

May 11 – Aug 21, 2026

- Built agentic developer workflows that orchestrate cellular ML model training, evaluation, Core ML packaging, and iPhone performance validation into reproducible end-to-end runs.
- Developed iOS **next-stage extensions of Call Context**, the feature announced at **WWDC26**, using on-device LLMs; integrated cross-functional internal infrastructure and managed session/prewarm, runtime invocation, and iPhone-side validation.
- Contributed upstream fixes/features to Apple ML tooling: **coremltools** LightGBM conversion [PR #2731] and **MLX** custom Metal kernel export/tracing support [PR #3650].

Open Source Contributor — MLC-LLM & Apache TVM

Embedding serving and compiler/runtime support | Python, C++, TVM

Remote

Feb 2026 – Present

- Contributed to **MLC-LLM** embedding serving/runtime across request scheduling, FFI plumbing, batch prefill, pooling, runtime guards, and Python/C++ parity tests.
- Upstreamed **Apache TVM** dependencies for masked sequence prefill, covering right-padded encoder batches with valid lengths and causal left-padding for decoder-only workloads.

Research Engineer (AI Agent)

Carnegie Mellon University | Python, FastAPI, Transformers

Pittsburgh, PA

Jul 2025 – Dec 2025

- Built a biomedical literature analysis platform with parallel retrieval from **PubMed** and **MedRxiv**, supporting Quick (<30s) and Deep (≤3min) structured analysis modes.
- Implemented embedding-based semantic search, reranking, structured report generation, and API workflows for extracting clinical variables and outcomes from research papers.

Machine Learning Research Intern

Lab of Cognitive & Computational Neuroscience, Santa Clara University | Python, TensorFlow, Shell

Santa Clara, CA

Jun 2024 – Sep 2024

- Developed CNN models and data-processing pipelines for neuroscience experiments; automated fMRI preprocessing on an HPC platform with Linux/Shell tooling.

Projects

LocalMelo — Localized AI Agent Framework | *Python, FastAPI, LLM Serving* [GitHub]

2026

- Designed **LocalMelo**, a **local-first** agent framework that separates LLM serving support from the agent runtime, with **planner**, **memory**, **executor**, and **sleep** components for localized AI workflows.

Medical Literature Analyzer | *Python, FastAPI, RAG, Transformers, Faiss* [GitHub]

2025

- Built an AI-powered literature analysis system with retrieval, semantic search, reranking, and report generation for clinical research workflows.

Food Recognition iOS App | *Swift, SwiftUI, TensorFlow, CoreML, Python* [GitHub]

2024

- Built a SwiftUI iOS app using MVVM and a TensorFlow-trained CoreML classifier; achieved 85% accuracy across 100+ categories and optimized inference by 4×.

MasumiRanker — AI Agent Discovery Platform | *Python, FastAPI, SQLite, Faiss, React* [GitHub]

2025

- Developed semantic agent search with Sentence Transformers and Faiss; built FastAPI/SQLAlchemy services for ratings, recommendations, and review integrity checks.